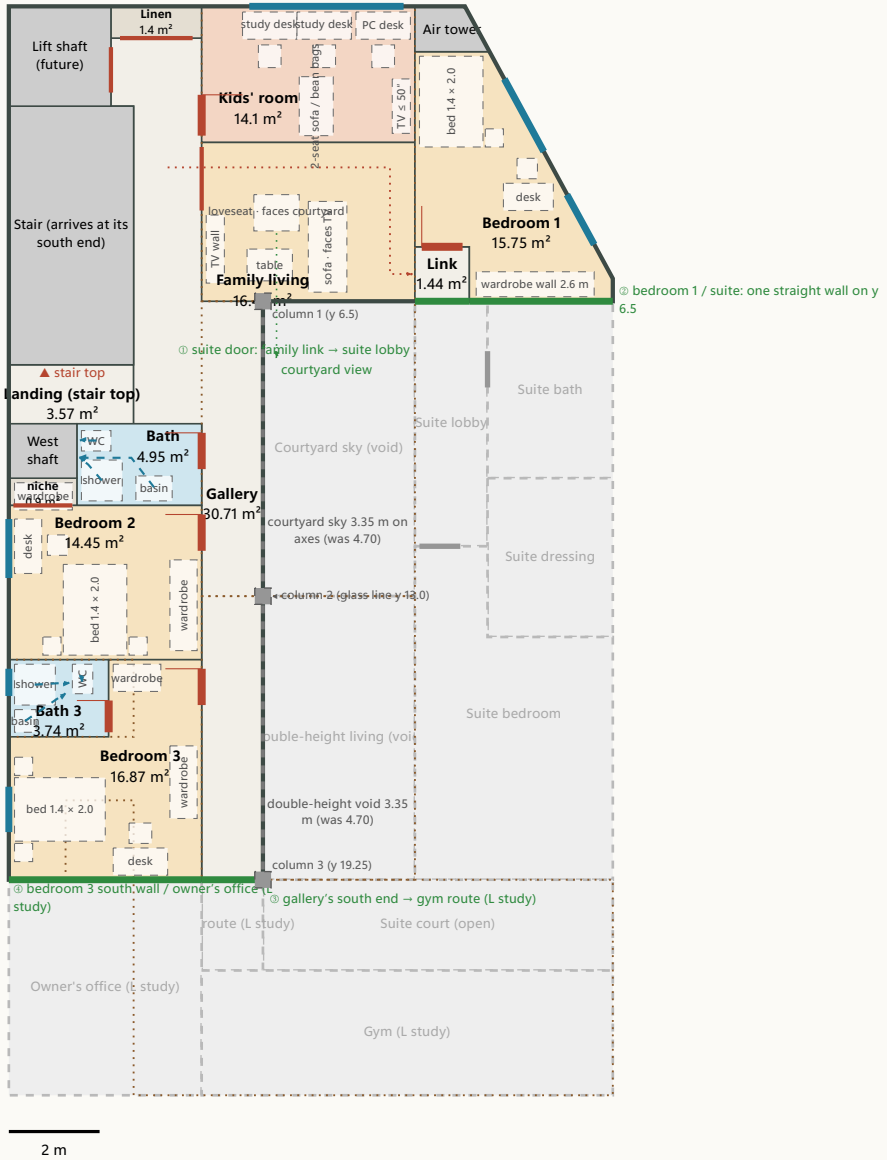
First floor · family part · focus study

Taif / Palm House · first floor V5.0 · 20 Oct 2026 · local = main CSV coordinates (x east, y south) · wall centre lines · north up

First floor V5.0 · part 1 of 2: family part · west projection 1.35 m, west rooms 4.25 m wide · part 2 (office, gym, suite court, master suite): studies/ff_suite.typ



Rooms

Room	Axes w × d (m)	Axes m²	≈ clear m²
Landing (stair top)	2.75 × 1.3	3.57	3.14
Gallery (north part + on the projection)	polygon	30.71	27.65
Linen cupboard	2 × 0.7	1.4	–
Family bath (bed 1 + 2, family living)	2.75 × 1.8	4.95	4.04
Wardrobe niche (bed 2)	1.5 × 0.6	0.9	–
Bedroom 2	4.25 × 3.4	14.45	12.84
Bedroom 3 bath (en-suite)	2.2 × 1.7	3.74	2.92
Bedroom 3	polygon	16.87	14.97
Kids' room (study + gaming)	4.7 × 3	14.1	12.44
Family living	4.7 × 3.5	16.45	15.16
Family link	1.2 × 1.2	1.44	1.08
Bedroom 1	polygon	15.75	13.83
Fixed: stair, lift, west shaft, air tower		23.8	
Zone		**148.14**	

Clear ≈ axes minus half a wall on every edge (250 mm outer, 200 mm inner, ≈ 50 mm balustrade on the void edges, nothing on open edges), corners ignored.

✓ All rooms inside the zone · no overlaps · no unassigned area · furniture inside its room, not overlapping, out of the routes · door swings clear · every room has a door reached from circulation · routes ≥ 1.0 m clear · suite door and gallery end on circulation

Key numbers

- **West projection** 1.35 m (x 4.25–5.6, y 6.5–19.25) on an edge beam and three columns: the gallery moves onto it and the west rooms take the whole 4.25 m band (≈ 4.03 m clear, was 2.53).
- **Courtyard sky** 4.70 → **3.35 m** on axes (≈ 4.50 → 3.30 clear); **double-height void** 4.70 → **3.35 m**. The ground-floor courtyard and living keep their full floor.
- **Gallery on the projection:** 1.35 m, ≈ **1.23 m clear** (½ wall + ½ railing). **Circulation in the zone:** 35.73 m² (landing, gallery incl. the projection, family link); the walk through the living is not counted.
- **Family living:** 16.45 m², on the courtyard edge, no separate route. **Kids' room:** 14.1 m².
- **Bedrooms:** 2 · 14.45 m² (4.25 × 3.4) · 3 · 16.87 m² + its own bath 3.74 m² · 1 · 15.75 m² (no bath). **Family bath** 4.95 m² for bedrooms 1 and 2, the family living and the kids' room.
- **Walked to the family bath door:** bedroom 2 ≈ 2.7 m · family living ≈ 8.8 m · bedroom 1 ≈ 17.3 m. Bedroom 3 → its own bath ≈ 2.7 m.
- **Owner's office:** bedroom 3 has moved up into the west band; the freed south-west corner (≈ 16 m²) becomes the owner's office, in the L study (studies/ff_suite.typ), its door on the gym route.

Plan: red = doors (thin = opening without a leaf) with swing line · blue = windows · blue dashed = drains · green = interfaces · brown dotted = ground floor below (the full courtyard and living floors) · grey dashed = the edge beam (x 5.6) and the three ground-floor columns · grey = neighbours · dashed boxes = furniture and fixtures.

Family part · the west projection and the rooms

West projection · 1.35 m over the courtyard and the double-height void

- **Where:** along the west edge of the courtyard and of the double-height living, x 4.25–5.6, from the courtyard's north edge (y 6.5) to the living's south wall (y 19.25). **West side only:** only the west rooms were narrow; a second one on the east would leave a slot of sky; the west projection shades the courtyard from the hard afternoon sun while the morning sun from the east stays; the suite on the east keeps its privacy.
- **Carried on:** an edge beam along x 5.6 and three columns, each in an existing wall or glass line, none free-standing in a room: **column 1** in the women's majlis south wall (y 6.5), **column 2** a pier in the glass between courtyard and living (y 13.0), **column 3** in the living's south wall (y 19.25). On the ground floor this reads as a colonnade (riwaq with columns) along the courtyard. **Alternative:** a plain 1.35 m cantilever from the x 4.25 line, with no columns and no edge beam (engineer to choose).
- **Before / after:** courtyard sky 4.70 → 3.35 m on axes (≈ 4.50 → 3.30 clear); double-height void 4.70 → 3.35 m; the ground floor under the projection (x 4.25–5.6) becomes normal height. The courtyard becomes deeper than wide (≈ 7–8 m walls / 3.3 m): less sky on its floor, more shade.
- **Gallery on it:** 1.35 m (≈ 1.23 m clear) with a railing (or low parapet with a screen) on the void edge; every west room door opens from it; no room opens onto the stair landing.

Kids' room · family living · bedroom 1 (unchanged)

- **Kids' room** 4.7 × 3.0 m: two study desks + PC desk under the north window, gaming corner (TV ≤ 50″ on the east wall, ≈ 1.3 m viewing distance), solid door from the gallery.
- **Family living** 4.7 × 3.5 m: a view, not a path; the walk to the family link passes behind and beside the sofa.
- **Bedroom 1:** one room, no bath, its south wall straight on y 6.5; it uses the family bath.

Family bath · 2.75 × 1.8 m, beside the west shaft

- WC on the shaft wall, shower 0.9 × 0.9, basin by the door; door from the gallery on the projection (opening in, swing clear), not onto the landing. For bedrooms 1 and 2, the family living and the kids' room.

Bedroom 2 · 4.25 × 3.4 m (≈ 4.0 × 3.2 clear) + wardrobe niche

- **Bed 1.4 × 2.0** with its head on the south wall and **bedside tables on both sides**; **wardrobes** in the niche beside the shaft and on the east wall (1.6 m); **desk** under the west window. Door from the gallery (swing clear), ≈ 2.7 m from the family bath.

Bedroom 3 · 16.87 m² + its own bath 2.2 × 1.7 m

- **Now in the west band** (it was on the men's majlis roof): an entry with a wardrobe, then the room 4.25 × 3.15 m: **bed 1.4 × 2.0** head on the west wall with two bedside tables, wardrobe on the east wall, desk on the south wall. Door from the gallery (swing clear).
- **Own bath (en-suite)** off its entry: shower, WC, basin; door swing clear; ≈ 2.7 m from the bedroom door. It stands **exactly over the GF family WC** (walls on the GF lines y 14.4 and 16.1).

Gym route and the owner's office → the L study

- The gallery ends at y 19.25 (x 4.25–5.6); from there the **gym route** and the **owner's office** (old bedroom 3 corner, ≈ 16 m²) belong to the L study (studies/ff_suite.typ), with the gym, the court and the suite.

Interfaces with the master suite study (fixed)

- ① suite door at the end of the family link (y 6.5); ② bedroom 1 / suite: one straight wall on y 6.5; ③ the gallery's south end (y 19.25, x 4.25–5.6), where the gym route starts; ④ bedroom 3's south wall (y 19.25), now a wall to the owner's office (no window).

Family part · routes, walls, drains, changes, open items

Routes (≥ 1.0 m clear)

Route	Axes	≈ clear	Deducted
Landing	1.3	1.2	½ wall on the south side; the north side is the stair arrival
Gallery north (beside the stair)	1.5	1.3	½ wall each side
Gallery on the projection	1.35	1.22	½ wall (rooms side) + ½ railing (void side)
Walk behind the sofa (in the living)	1.1	1	½ wall on the kids' room side; the sofa side is open
Walk beside the sofa to the family link	1.1	1	½ wall on the bedroom 1 side; the sofa side is open
Family link	1.2	1	½ wall each side (bedroom 1 to the north and east)

Walking distance	m
Stair top → kids' room door	8.8
Stair top → family living (opening)	8.7
Stair top → bedroom 1 door	15.9
Stair top → ⓘ suite door	15.9
Stair top → family bath door	5.7
Stair top → bedroom 2 door	7.5
Stair top → bedroom 3 door	10.9
Stair top → gallery's south end (L)	14.5
Bedroom 2 door → family bath door	2.7
Family living → family bath door	8.8
Bedroom 3 door → its own bath door	2.7
Bedroom 1 door → family bath door	17.3

Stair top → every room, the suite door and the gallery's south end runs only on the landing, the gallery (north part and on the projection), the family link and the walking line through the living (checked: every door's approach is on circulation, no furniture in a route).

Engineer

- **Projection:** 1.35 m slab beyond x 4.25 along y 6.5–19.25; edge beam on x 5.6 (spans ≈ 6.5 m and 6.25 m, depth to confirm; seen from the courtyard and the living, best expressed as the colonnade's beam); three columns down to their own foundations (sizes and pads to confirm). Or a plain cantilever without columns.
- **Ground floor (frozen V4.0) changes only by:** the three columns (two inside walls, one a pier in the courtyard / living glass at y 13.0), their foundations, the beam line and the lower ceiling under the projection.

Walls off the ground-floor lines

Ground-floor lines here: x 0, 1.25 (guest WC), 1.5 (shaft), 2.25, 2.75, 4.25, 8.95 · y 2.2, 7.9, 10.4, 12.9, 14.4, 16.1, 17.5, 19.25, 21.4 (foyer) (west band); the women's majlis (x 4.25–8.95, y 0–6.5) and the men's majlis (x 2.75–13.311, y 19.25–24) have no inner wall; east part (bedroom 1, family link): see the V4.1 list.

- **Off:** y 3.0 (kids' room / living, over the women's majlis); y 11.0 (family bath and niche / bedroom 2) and x 1.5 south of the shaft (niche); x 2.2 (bedroom 3 bath / entry); (the gym route walls: see the L study).
- **On:** y 14.4 and 16.1 (bedroom 3 bath = the GF family WC walls), x 1.5 / y 10.4 (shaft), y 6.5 (bedroom 1 / suite), **x 4.25** (the rooms' east wall, on the GF courtyard edge line), the outer walls.
- **Gone upstairs:** the wall on x 2.75 south of the landing (the rooms now span over the GF riwaq); the GF riwaq wall x 2.75 below carries the floor as before.

Drains (blue dashed)

- **Family bath:** WC on the west shaft wall (≈ 0.3 m), shower ≈ 0.5 m, basin ≈ 1.8 m, in the bath floor to the shaft; floor over the GF utility room and hospitality kitchen, nothing in their ceilings.
- **Bedroom 3 bath:** exactly over the GF family WC: WC straight into its stack, shower ≈ 0.9 m, basin ≈ 1.1 m. Wet room on a wet room.
- **Bedroom 1:** no bath. The east duct is removed (suite bath to a stack outside the east wall, see ff_suite.typ).

What changed in this step (19 Oct 2026)

- **West projection 1.35 m** with the gallery on it; west rooms 4.25 m wide; courtyard sky and void 3.35 m.
- **Bedroom 2** 4.25 × 3.4 m; **bedroom 3 + own bath** moved up into the west band; **family bath** by the shaft, door from the gallery.
- **Gym route** moved to the end of the projection; the old bedroom 3 corner is freed; the suite court loses 1.35 m at its west end.
- **Reverses** the 18 Oct decision that the gallery stays above the GF riwaq (x 2.75): the rooms now span over the riwaq and the gallery is on the projection.

Open items

- **Built-in cupboards in niches** (linen cupboard 2.0 m, bedroom 2 wardrobe niche 1.5 m): both already have walls on both sides. Prefer a **joinery box** (own 18 mm sides, top and frame, scribed to the walls): no block, full width kept. Use a **10 cm block jamb + header** only where a side is open or a heavy sliding door needs a solid fixing, and then draw the niche 20 cm wider so the clear width stays. Decide with the carpenter (docs/DETAILS.md).
- **Owner's office** decided (19 Oct 2026): the freed south-west corner, in the L study.
- **Columns or cantilever** for the projection (engineer + owner: the colonnade is an aesthetic choice).
- **Courtyard light:** sky 3.3 m between ≈ 7–8 m walls; to check the courtyard and the double-height living with a daylight study.
- **East duct removed:** the ground floor (frozen V4.0) still shows it; to take out of design_gf.typ once approved.